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Inventor(s)	Neely; Jared et al.

Morphing airfoil

Abstract

The present disclosure defines a morphing airfoil having a dynamic flexible skin system that is capable of carrying high level aerodynamic (or fluid) pressure loads over a structural surface. The structural surface can morph and deflect in response to control inputs to change a lift force without separate movable control surfaces. The anisotropic skin is attached to underlying structure that is both actively controlled and passively supported. A control system causes the underlying support structure to move to a desired location which in turn causes the skin to bend and/or flex without exceeding a stress threshold and thus vary the lift profile along a span of the airfoil.

Inventors: Neely; Jared (Troy, OH), Joo; James (Centerville, OH)

Applicant: The Government of the United States as represented by the Secretary of the Air Force (Wright-Patterson AFB, OH)

Family ID: 84324929

Assignee: United States of America as represented by the Secretary of the Air Force (Wright-Patterson AFB, OH)

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Primary Examiner: Berona; Kimberly S

Assistant Examiner: Aleksic; Nevena

Attorney, Agent or Firm: AFMCLO/JAZ

Background/Summary

(1) Pursuant to 37 C.F.R. § 1.78(a)(4), this application is a Continuation of co-pending patent application Ser. No. 17/130,448 filed Dec. 22, 2020 which claims the benefit of and priority to prior filed Provisional Application Ser. No. 62/957,499, filed Jan. 6, 2020, now expired, which is expressly incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure generally relates to a morphing airfoil (or hydrofoil) having a non-stretchable flexible anisotropic load carrying skin with underlying structure that includes active control elements and passive elements that change the camber of the airfoil.

BACKGROUND

(2) Airfoils provide aerodynamic or hydrodynamic lift and control for vehicles such as aircraft, watercraft, automobiles or other apparatus that move through a fluid. Airfoils typically include a movable portion operably connected to a fixed portion to change the variable lift forces across a span thereof. As an example, aircraft wings and tails include fixed or static structure with connected movable portions such as ailerons, flaps, slats, rudders, elevators, etc. The movable

portion of the airfoils are used to vary the lift of the airfoil and therefore permit maneuverability of the vehicle within an ambient fluid. By using only a portion of an airfoil to vary the lift, inefficiencies in the form of pressure (drag) losses are generated.

(3) When industry and academia have looked at morphing aircraft wings in the past, they have focused primarily on the structure that creates the shape change under the skin of the wing. The problem is that to achieve the complex surface geometry the skin needs to be able to bend and flex across the entire wing in addition to carrying aerodynamic pressure loads. Elastomers are a common material used in morphing skin concepts, however they tend to be heavy and susceptible to UV decay. To withstand the aerodynamic pressure loads, the elastomer must be pre-tensioned to be effective. The use of elastomers can lead to skin drooping or ballooning under high pressure loads, and the high strains associated with deforming a pre-strained skin can lead to high actuation loads and undesirable energy storage in the elastomeric skin. As these and other existing systems have various shortcomings, drawbacks, and disadvantages relative to certain applications, there remains a need for further contributions in this area of technology.

SUMMARY

(4) One embodiment of the present disclosure includes a unique morphing airfoil for use on vehicles that move through a fluid. Other embodiments include apparatuses, systems, devices, hardware, methods, and combinations wherein the morphing airfoil includes a non-stretchable skin that is flexible and controlled to a desired shape to provide for aerodynamically efficient lift and control of the vehicle. Further embodiments, forms, features, aspects, benefits, and advantages of the present application shall become apparent from the description and figures provided herewith.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIG. 1 shows a section of compliant skin material with a number of standoff mounts attached thereto and with stringers extending through the mounts;
- (2) FIG. 2 is an enlarged view of a mount and stringer from FIG. 1;
- (3) FIGS. 3A-3D are perspective views of exemplary standoff mounts according to some embodiments of the present disclosure;
- (4) FIG. 4 is a cross-sectional partially transparent view of a cambered wing having standoff mounts attached to the compliant skin and flexible stringers placed through the mounts;
- (5) FIG. 5A is another cross-sectional view of a cambered wing section having standoff mounts attached to the compliant skin and stringers placed through the mounts;
- (6) FIG. 5B is a side-view of a cambered wing section having standoff mounts attached to the compliant skin and stringers placed through the mounts;
- (7) FIG. 6 is a bottom plan view of a cambered wing section with a portion of the skin removed to show standoff mounts and stringers attached thereto;
- (8) FIG. 7 is another cross-sectional view of a cambered wing section having standoff mounts attached to the compliant skin and stringers placed through the mounts;
- (9) FIGS. 8A-8C are side, perspective and cross-sectional views of a compliant structural support rib;
- (10) FIGS. 9A and 9B are side cross sectional views of the compliant structural support rib in multiple deflected configurations;
- (11) FIGS. 10A-10C are perspective views of a main actuator rib;
- (12) FIGS. 11A-11F are various views of the main actuator rib structure in deflected positions;
- (13) FIG. 12 is a perspective view of a portion of a morphing cambered wing;
- (14) FIG. 13 is a view of a wing structure with the flexible conforming skin removed to reveal internal parts;

(15) FIGS. 14A, 14B, and 14C are various views of the front spar support located proximate the leading edge of an airfoil;

(16) FIG. 15 is another view of a complete wing structure with a portion of the skin cut away to show internal components; and

(17) FIG. 16 is another view of a complete wing structure with a portion of the skin cut away to show internal components.

DETAILED DESCRIPTION OF THE ILLUSTRATIVE EMBODIMENTS

(18) For the purposes of promoting an understanding of the principles of the invention, reference will now be made to the embodiments illustrated in the drawings and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the invention is thereby intended. Any alterations and further modifications in the described embodiments, and any further applications of the principles of the invention as described herein are contemplated as would normally occur to one skilled in the art to which the invention relates.

(19) The present disclosure is directed to an airfoil having flexible/compliant skin that has anisotropic mechanical properties that are necessary to maintain a surface shape thereof. The skin carries the fluid dynamic pressure loads across a morphing/deforming control surface and distributes the net pressure loads to the underlying structure. It should be noted that while the disclosed embodiment is directed to an aircraft wing that “morphs” into different shapes to change lift and drag characteristics thereof, the present invention is not limited as such. For example the morphing airfoil may be used in other applications such as automotive or watercraft operations. Terms describing the morphing airfoil or portions thereof can include control surfaces, wings, tails, airfoils, hydrofoils, waterfoils and as such may be used interchangeably throughout the specification. Regardless of which terms are used herein, each denote fluid dynamic structures that generate lift forces and variation in lift forces as a vehicle moves through a fluid or in stationary structures such as windmills.

(20) The flexible/compliant skin design for camber morphing control surfaces addresses the need for a load carrying skin that can be used to create a smooth and continuous control surface profile, which can be used to span large distances between the control points (actuator ribs) that drive surface deformation. The representative embodiment described herein allows the underlying surface control structure to interface with the skin in a manner that transfers loads without using traditional fastening and bonding techniques. Prior techniques that bonded the skin to the structure are more highly constrained in degrees-of-freedom of the skin-to-structure interface. Structure-bonded skin also results in high actuation forces and undesirable interactions between internal control actuators. This occurs when actuated control points are independently controlled and differ in deflection angle because the distance between the deflected control points, having differing deflections, will always be larger than the distance between deflected control points which deflect in unison (if no skin is bonded). However, if a non-elastic material is bonded across the control surface, the distance between the control points is fixed, and the control points are forced to move/bend inward towards each other to maintain the same fixed distance between control endpoints, while in the presence of differing deflections. The adaptive conformal skin system eliminates the aforementioned cause of actuator interaction by allowing the actuated control points (actuated ribs) to slide relative to the underside of the fixed length (non-elastic) skin surface spanning across the structure.

(21) The flexible/compliant skin design can be implemented into any application where a morphing/deforming control surface in a fluid flow is desired to improve control performance and reduce drag. One such application is in the aerospace industry, where aircraft wings, elevators, rudders, empennage structure, shape changing air intakes, propeller blades, or other lift surfaces/devices are used. Another application is in the automotive industry where shape changing spoilers or exterior flow control can improve performance. Yet another application is the marine industry where a smooth surface camber changing rudder or other control surface can improve

operating performance.

(22) The present invention addresses the need for a morphing control surface skin that secures to (and conforms to) a surface shape defined by multiple internal actuated structural members, without adding unnecessary actuation forces to the overall morphing control surface system. This arrangement is permissive of pivoting and translation relative to the underlying dynamic structure by allowing degrees of freedom which would otherwise contribute to an undesirable in-plane shear stress within the deforming skin.

(23) The flexible composite laminate skin is the component that comes into direct contact with the fluid flow on the airfoil surface and creates a smooth lifting control surface. In one embodiment, a layup of composite fiber/fabric plys can be used with unidirectional fiber/fabric running in the spanwise direction, in order to provide additional stiffness for carrying skin pressure loads across unsupported spans between surface deformation control ribs or similar control mechanisms. Additional layers of fabric can be placed in areas where additional stiffness is desired, such as along the trailing edges of the skin.

(24) Flexible stringers can be formed from unidirectional carbon fiber extruded rods or tubes. In one embodiment, the flexible stringer have a square cross-section, but other cross sectional configurations are contemplated herein. The stringers can be positioned parallel to the spanwise direction of the airfoil or wing. The stringers carry the aerodynamic loads from the skin, transferred through stringer standoff mounts, and distribute the forces across the static and dynamic internal structural components of the airfoil.

(25) Stringer standoff mounts are bonded or fastened to the inner surface of the skin. The standoff mounts provide an elevated connection between the flexible stringers to the skin. Spacing the stringers away from the inner surface of the skin increases the out-of-plane moment of inertia, which serves to further increase out-of-plane and spanwise bending stiffness of the airfoil. The standoff mount configuration also provides space between the inner skin surface and the stringers, which is used to interface the stringers with the internal structure and actuated shape control devices. Standoff mounts are positioned to the internal side of internal structural components, for attaching and constraining motion. The standoff mounts can be spaced so that the distance between the underlying structure and/or standoff mounts is generally consistent across all the structure. In other forms the mounts are positioned at variable distances from one another along a length of the airfoil. The mounts are also placed close enough together to sufficiently distribute the pressure loads to the stringers.

(26) The stringer standoff mounts have a unique pass through opening that allows the flexible stringers to pivot inside the standoff mounts. The pass-through opening has an hourglass shape in some embodiments. This hourglass shape reduces stress concentrations on the stringers at the edge of the standoff mounts, particularly when the skin is simultaneously bending and twisting. This allows the stringers to bend and twist with the flexible composite skin. To further increase the flexibility of the stringer reinforced skin, the stringers are not constrained in a longitudinal direction such that sliding can occur relative to the pass-through of the standoff mounts when the airfoil morphs into different shapes. This is achieved by fixing all degrees-of-freedom of the standoff mount-stringer interface at a single location for any given stringer. Redundant fastening mechanisms are installed to prevent stringer slide-out, should the single fixed location fail during operation.

(27) A dynamic conformal skin system provides for camber morphing control surfaces that are smooth and continuous (e.g. without flow interruption features such as hinges, slots and the like). The control surface profile can span large distances between the control points that drive surface deformation. The representative design configuration allows the underlying surface control structure to interface with the skin in a manner that transfers loads without using traditional fastening and bonding techniques. Such prior fastening or bonding techniques have been shown to constrain the degrees-of-freedom of the skin-to-structure interface and can result in high actuation

forces and undesirable interaction between internal control actuators.

(28) The dynamic conformal skin system has the potential to be implemented into any field-of-use where a camber morphing/deforming control surface in a fluid flow is desired. The deformable surface can serve to improve overall control performance and reduce drag in the implemented device. One application is in the aerospace industry, for use on parts/surfaces such as aircraft wings, elevators, rudders, empennages, shape changing air intakes, propeller blades, other lift surfaces/devices, and so forth. The dynamic conformal skin system can also be used in the automotive industry for shape changing spoilers or exterior flow controlling surface panels. The dynamic conformal skin system can also be used in the marine industry, for instance as a smooth surface camber changing rudder or other type control surface that would benefit from a smooth surface.

(29) The dynamic conformal skin system is a novel approach to creating a camber morphing control surface, which is accomplished by combining a flexible dynamic outer skin that translates and conforms to an underlying shape controlling structure. The underlying structure is comprised of multiple active and passive shape control components to achieve a smooth and continuous control surface.

(30) The flexible anisotropic load carrying skin for camber morphing control surfaces, which is used in this camber morphing control system is a part of the overall system. The flexible anisotropic load carrying skin is composed of a flexible composite skin, flexible stringers and stringer standoff mounts. The compliant structural support ribs for camber morphing control surfaces, which a part used in this camber morphing control system is described in more detail above.

(31) The dynamic skin system is capable of carrying design limit level aerodynamic pressure loads in a manner wherein the flexible stringers interface with the actuated position control ribs, the passive compliant support ribs, and the flexible composite skin system. The implemented system is equipped with a number of strategically positioned stringer standoff mounts which receive the flexible stringers and help to control the bending of the skin and structural surface.

(32) Referring now to FIG. 1, a portion of an airfoil **100** having a section of flexible compliant skin **102** with standoff mounts **104** attached thereto is depicted. The skin **102** can generally be any shape or configuration depending upon the application for the airfoil **100**. The standoff mounts **104** are attached via mechanical means including a glue or an epoxy to an inner surface **108** of the skin **102** in a plurality of locations. A plurality of stringers **106** are inserted through openings **110** in the standoff mounts **104** lengthwise or longitudinally along the surface **108** of the skin **102**. The stringers **106** transmit an actuation force to the skin **102** such that the skin **102** morphs into a different shape to change the lift profile of the airfoil without a separate hinged control surface as is traditional.

(33) Referring now to FIG. 2, an enlarged view of a standoff mount **104** is illustrated. The mount **104** includes an opening **110** extending therethrough that has an hourglass shape e.g. inwardly tapered wall extending from either end thereof. This allows the stringer **106** to slide through the opening, but also allows the mount to twist slightly without bending the stringer, wherein such bending might cause problematic stresses on the skin **102**, the mount **104**, and the stringer **106**.

(34) Referring now to FIGS. 3A through 3D, several views of representative standoff mounts **104** are shown. The standoff mounts are designed to be attached (glued or otherwise mechanically affixed) to the inner surface of the flexible skin **102**. The stringers **106** extend through the openings **110** in the mounts **104** at a defined distance from the skin surface. The standoff distance can vary from a relatively small distance to a relatively large distance, depending upon the system application.

(35) Referring more particularly to FIGS. 3A and 3B, perspective views showing one embodiment of a standoff **104a** with solid and hidden lines, respectively. The standoff **104a** generally includes bottom mounting surface **120a** that is configured to be uneven (for instance, rough, waffled, scored,

or castellated). This uneven bottom surface **120a** can assist in attaching the standoff mount **104a** to the surface of the compliant skin **102**. The mount **104a** includes opposing sidewalls **124** with front and back walls **126**, **128**. An opening **110** is formed through the front wall **126** and extends through the back wall **128** and is configured to receive a stringer **106** therethrough. It should be noted that the mounts **104** may have shapes other than that shown in the disclosed embodiments. For example, the mounts **104** may be round or may include more than four distinct outer walls. In the disclosed embodiment, the opening **110** is shown to be square and has tapered inner sides **132**, which form an hourglass shape when viewed from above in a cross-sectional view. The hourglass opening **110** includes tapered or angled sidewalls **132** extending from the front and back walls **126**, **128**, respectively and intersect at an intermediate position internal to the mount **104**. These tapered walls **132** allow the stringer **106** to run through the opening **110** without undue stresses being exerted upon the stringer **106** and the mount **104** and the various parts move to effectuate bending of the associated underlying structure. It should be noted that other configurations of openings (e.g. other than hourglass) **110** may be used without departing from the teachings of this disclosure.

(36) FIGS. 3C and 3D depict an alternate embodiment of a mount **104b** with a solid perspective view and a hidden line perspective view, respectively. The mount **104b** is similar to mount **104a** with the exception that it has a substantially flat bottom **120b**.

(37) FIG. 4 is a perspective side view of an exemplary cambered wing section **150** with the skin removed to show standoff mounts **104** and stringers **106** along a trailing edge portion **162**. The bottom surface and the top surface of the flexible conforming skin (removed for clarity) of the trailing edge portion **162** includes a series of standoff mounts **104** attached thereto. The upper and lower standoff mounts **104** have stringers **106** that are positioned through the series of openings created by the (generally) aligned mounts **104**. A translational lap joint **152** is located on the lower skin surface that includes overlapping skin that permits translation and pivoting motion when the wing is deformed or morphed.

(38) FIGS. 5A and 5B shows other views of a cambered wing structure **100** having a conforming flexible skin **102**, standoff mounts **104**, and stringers **106** placed through openings **110** in the mounts **104**. A series of standoff mounts **104** are attached to the lower portion **164** of the flexible conforming skin **102**. The stringers **106** are positioned through the openings **110** and held by the attached mounts **104**. Similarly, a series of standoff mounts **104** are attached to the upper portion **166** of the flexible conforming skin **102** to receive stringers **106** connected therethrough along a trailing edge **162** of the wing structure **100**.

(39) FIG. 6 is a bottom plan view with the skin removed to show hidden lines of a cambered wing section **100**. The trailing edge **162** of the cambered wing **100** is the portion that is configured to move or bend. Therefore, a series of standoff mounts **104** are strategically attached along the surface of the flexible conforming skin **102**. A series of stringers **106** are then positioned through aligned openings **110** of the attached standoff mounts **104**. A translational lap joint **152** is located on the lower skin surface that includes overlapping skin that permits translation and pivoting motion when the wing is deformed or morphed.

(40) FIG. 7 is another view of a cambered wing section **100** having a flexible conforming skin **102** with standoff mounts **104** affixed thereto and stringers **106** running through the openings **110** in the standoff mounts on the suction side **166** and the pressure side **164** of the wing **100**. A translational lap joint **152** is located on the lower skin surface.

(41) Referring to FIGS. 8A-8C, a compliant structural support rib **200** is illustrated in a side view and a perspective view, respectively. A plurality of compliant ribs **200** are assembled within the morphing airfoil to provide structural support while permitting a morphing movement of the airfoil. The compliant ribs **200** provides structural capabilities that maintain the cross-sectional airfoil profile through a wide range of control surface deflections between actuated control points.

(42) The purpose of the compliant structural support ribs **200** is to maintain and control the airfoil profile of a morphing control surface in-between actuated (active) ribs (to be described below),

where the upper and lower control surface skins would otherwise be unsupported between active ribs. The compliant support ribs **200** constrain the translational motion of the control surface skins relative to one another. The support ribs **200** enable effective and efficient transfer of pressure loads into the primary load-carrying structure through a full range of dynamic motion, while adding minimal actuation force resistance to the morphing control surface system.

(43) In the exemplary embodiment, the compliant support ribs **200** have a unique square wave shaped structure **202** that is formed between the upper and lower control surfaces to provide support between the pressure side and suction side surfaces of the airfoil under loading and during actuated deflection (e.g. morphing). The compliant support ribs **200** have integrated stringer channels or slots **210** and **212** to constrain the motion of stringers **106** (not shown) that extend therethrough. Wall thicknesses **214** at the forward end and **215** at the aft end of the compliant support ribs **200** are optimized to bend and deflect to the target deformed airfoil profile with minimal applied force to reduce the required control surface actuation force, while providing sufficient structural support of the skin. The wall thicknesses **214** and **215** may be the same in some embodiments and different in others.

(44) The compliant support ribs **200** are passive dynamic structural devices that are driven from the displacement of the flexible skin and flexible offset stringers. The compliant support ribs **200** maintain the airfoil cross-section between actuation ribs that are of equal deflection and transition the deflection of the airfoil cross-section when located between actuated control points of differing deflection angles. This results in a smooth and continuous transition between actuator ribs, which creates a surface that reduces flow separation at the trailing edge **216**.

(45) In one embodiment, a square, rectangular or other periodic wave shape structure **202** forms the structural shape of the rib **200**. The trailing edge **216** of the support rib **200** is movable in an up and down manner to morph in response to movement of the skin when the airfoil actuators are activated as will be described in more detail below. The compliant rib **200** includes a pair of opposing arms **230** extending forward proximate a leading edge of the airfoil. Each of the opposing arms **230** project radially inward from a first portion and terminate at a forward facing lip **232**. A through aperture **234** is formed through each of the lips **232** to facilitate clamping or fastening around a front or main spar (not shown). An enclosed element **240** extends aft of the opposing arms **230**. The enclosed element **240** includes a forward wall **242** and an aft wall **244** with an upper wall **246** and lower wall **248** extending between the forward and aft walls **242**, **244** respectively. A stringer channel **212** connected to the upper wall **246** and the lower wall **248** of the enclosed element **240**. The periodic structure **202** formed in a substantially square wave pattern extends aft of the enclosed element **240**. The periodic structure **202** includes intermittent bottom walls **250** connected to intermittent top walls **252** with side walls **254** extending therebetween. The top walls **252** and bottom walls **250** can be inwardly tapered toward the trailing edge **216** to provide support for skin in an airfoil shape. Each of the top walls **252** and the bottom walls **250** include a stringer slot **210** connected thereto. The periodic structure **202** is flexible, but provides sufficient strength to hold the skin in a desired shape. The trailing edge **216** is defined by a triangular structure **260** with a solid wall **262** formed therein.

(46) Referring more particularly to FIG. **8B**, a cross-sectional view of section A-A in FIG. **8A** illustrates a revolving hour-glass shaped stringer pass-through aperture **218** which permits a trailing edge stringer **800** (see FIG. **13**) to be connected through the trailing edge of each compliant support rib **200**, as well as through each actuated control rib to be described below. The stringers **106** pass through the motion constraining slots **210** for relatively smaller displacements and through constraining slots **212** for larger displacement control depending on the orientation of support rib installation. A surface at the trailing edge **216** is attached to the inner surface **108** (not shown, see FIG. **1**) of the skin **102** (not shown, see FIG. **5B**) by means bonding or traditional fasteners. This helps keep the shape of trailing edge with the help of the trailing edge stringer **800** (see FIG. **13**) through the hourglass cross-sectional holes **218** of compliant ribs **200** and the aperture **554** (see

FIG. 13) located near the trailing edge of the actuated ribs **400** (see **FIG. 13**).

(47) Referring now to **FIGS. 9A and 9B**, an exemplary airfoil **100** with a compliant support rib **200** is shown in deformed and un-deformed states. The un-deformed (neutral) state **302** of the compliant support rib **200** is shown along with maximum deflected upward position **304** and maximum downward deflected position **306**. The position of the trailing edge can be moved anywhere between the maximum upward and downward positions **304, 306** respectively. The deflections cause a continuous (no discrete hinge) parabolic flap airfoil profile. The flexible composite skin **102** forms a continuous morphed surface profile around the support ribs **200**. The compliant support ribs **200** are capable of achieving smooth deflections that follow the deflected conformal airfoil profiles in the upward deflected **304** and downward deflected **306** configurations.

(48) Referring now to **FIGS. 10A-10C**, a primary or actuator rib **400** is configured to provide dynamic control of the camber morphing of the conformal skin system. A plurality of actuator ribs **400** control the shape of the airfoil. Each actuator rib **400** includes a motion control slot **402** and a compliant mechanism **404** to permit flexible movement of the actuator rib **400** to achieve effective camber control. The compliant mechanism **404** provides smooth motion in a chordwise direction (double arrow **406**) due to the vertical flexible compliant legs **408** that bend under a chordwise force. An upper portion **409** of the actuator rib **400** is fixed to a front spar (not shown) via a pass through aperture **410**. The front spar (not shown) passes through the aperture **410** formed in a vertical support member **412** that is connected to the upper portion **409** of the actuator rib **400**. The motion of slot **402** is operable to slide back and forth across the aperture **410** and fixed front spar. A control linkage attachment point **466** (See **FIGS. 11A-11C**) transmits a force from a rotary drive actuator (not shown) into the actuator rib **400** at the aperture **463** to deflect the trailing edge **432** up and down. The rotary actuator (not shown) is mounted onto the actuator rib **400** using mounting holes **416**. The actuator rib **400** attaches to a main spar **500** (see **FIG. 12**) with a spar attachment member **418** which can be clamped together by way of a fastening flange **460**. Material can be removed from the actuator rib **400** for weight reduction at locations **420** and **422**. A tapered stiffness member **424** aft of the spar attachment member **418** on the upper surface **409** provides hinge point tailoring in combination with a compliant web **430**. These features permit the upper surface **409** and a lower surface **411** near the actuator rib **400** to be connected together in a flexible manner.

(49) A rigid web **434** extends between the upper and lower surfaces **409, 411** at the aft portion of the trailing edge **432**. This structure causes all features aft of the rigid web **434** to translate as a displacement extension of the forward compliant features.

(50) The compliant mechanism **404** has a split line **450** at an intermediate location that can be at the centerline in some forms. Similarly, the motion control slot **402** has a split line **452** at an intermediate position. The split lines **450, 452** allow the control linkage to be installed inside the rib **400** proximate the center plane. The triangular area **455** aft of the rigid web **434** is used to mount various inserts and connector interfaces with the trailing edge stringers (not shown), similar to the attachment feature **218** in the compliant support rib **200**.

(51) Referring now to **FIGS. 11A through 11F**, the actuator rib **400** is shown in side views and perspective views of neutral and deformed states (e.g. trailing edge **432** upwardly deflected and downwardly deflected.) In **FIGS. 11A and 11D**, the trailing edge **432** is deflected upward due to the motion control slot **402** being pulled in the direction of the main spar by the control linkage **465**, while sliding on the front spar **502** (not shown, see **FIG. 12**), and reaches the forward end of the motion control slot **402**. The interior surface of the motion control slot **402** slides on the front spar **502** (not shown, see **FIG. 12**) which passes through aperture **410** of the actuated rib **400**. The vertical compliant legs **408** of the compliant mechanism **404** and the compliant web **430** are deflected to permit the lower surface **411** to move forward relative to the upper surface **409** of the actuator rib **400** and thus causing the trailing edge to deflect upward. The actuator rib **400** is powered by a rotary actuator (not shown) with shaft axis of revolution at location **466**. The shaft of

the rotary actuator **466** turns a control arm and connects to a control linkage **465** at the pinned joint **467** of the control arm. A pin (not shown) is inserted through the aperture **463**, connecting the ball joint rod end of the control linkage **465**. The rotary drive actuator (not shown) transmits force into the actuator rib **400** at a pinned connection **463** through the control arm **464** and control linkage **465**.

(52) In FIGS. **11C** and **11F**, the trailing edge **432** is deflected downward due to the control arm/linkage **464,465** pulling the pinned connection **463** in the forward direction. The vertical compliant legs **408** of the compliant mechanism **404** and the compliant web **430** are deflected to permit the upper surface **409** to move aft relative to the lower surface **411** of the actuator rib **400** and thus causing the trailing edge to deflect downward. FIGS. **11B** and **11E** show a neutral condition—e.g. no actuation force acting on the actuator rib **400**.

(53) Referring now to FIG. **12**, a partial airfoil **100** cutaway to show a complete conformal skin system for a cambered morphing control surface is depicted. The complete system includes flexible conforming skin **102** positioned around the internal structural parts. The internal structural parts include a main spar **500** and a forward spar **502** configured to connect the actuator ribs **400** and the compliant ribs **200** together. The standoff mounts **104** are attached to the flexible conforming skin **102** and the stringers **106** are inserted through the openings in the mounts **104** and connects the conforming skin **102** to the actuator ribs **400** and the compliant support ribs **200**. The stringers **106** pass through slots **210** and **212** (FIG. **8A**) of the compliant support ribs **200** and across the inner surfaces **461,462** of the actuated ribs **400**. The leading edge structural support **600** attaches to the main spar **500**, by means of the flange **606** (see FIG. **14**), and makes **600** a cantilever member. A plurality of leading edge supports **600** supports the front spar **502** by attachment at aperture **602** (see FIG. **14**). The leading edge interior surface of the flexible conforming skin **102** is attached to the skin mating surface **608** (see FIG. **14**) of the plurality of leading edge supports **600**.

(54) Referring now to FIG. **13**, a portion of the airfoil **100** showing an inner system **550** with the skin removed for clarity is depicted. A plurality of primary actuation ribs **400** are intermittently spaced across the wing structure with compliant structural support ribs **200** intermittently positioned therebetween. A stringer **800** in the trailing edge **162** interfaces with the actuator ribs **400** using a revolving hour-glass shaped stringer pass-through aperture **552**, similar to that of **218** (see FIG. **8B**), in the trailing edge insert **554**. The trailing edge insert **554** is connected to the triangular region in the trailing edge of the actuated rib **400**, located aft of the rigid web **434** (see, FIG. **10**). The same trailing edge stringer **800** also interfaces with the compliant support ribs **200** through an aperture insert **218**. Stringers **106** pass-through the openings of stringer standoff mounts **104** to connect the ribs **400, 200** to the skin (not shown). The stringers **106** glide on the interior of the actuator ribs **400** on both the upper **166** and lower surfaces **164**. The stringers **106** are attached to the skin (not shown) via the standoff mounts **104** and slide within the motion containing slots **212, 210** of the compliant support ribs **200**.

(55) Referring now to FIGS. **14A** through **14C**, a leading edge structural support member **600** is illustrated in various views. The leading edge structural support member **600** is optimized to provide an optimal stiffness per unit volume to maintain the rigidity of the front spar **502** (not shown—See FIG. **12**) which interfaces through an aperture **602**. The leading edge is stiffened by a vertical web **604** at the mid-plane of the leading edge structural support member **600**. A flange **606** is connected to a main spar as illustrated above in FIG. **12**. A plurality of leading edge structural support members **600** are positioned at intermediate locations between and/or beside the actuator ribs **400**. The skin mating surface **608** is attached to the leading edge interior surface of the flexible conforming skin **102** (not shown, see FIG. **12**) by means of bonding or traditional fastening techniques.

(56) Referring now to FIGS. **15** and **16**, complete morphing wing sections **700** are shown with cutaways of portions of the conforming skin **102** in order to show the internal components of wing structure. The leading edge structural support members **600**, actuator ribs **400**, compliant ribs **200**,

stringers **106**, standoff attachments **104** and main spar **500** are shown in an assembled condition. Compliant support ribs **200** are shown spaced between the actuator ribs **400**. The standoff mounts **104** are shown distributed across the inner surface of the flexible conforming skin **102** with stringers **106** running through the openings in the standoff mounts.

(57) In one form, the composite laminate skin surface can be constructed from multiple carbon fiber plys or layers, including a combination of woven and/or unidirectional fabrics, giving the skin different properties in different directions (anisotropic). For instance, the composite skin can be made using a dry fabric layup on a profiled tool and a resin infusion process.

(58) As an example, the cured composite skin can be deburred and trimmed to the desired dimensions. Locations for stringer offset mounts can be indexed and marked on an inner surface of the flexible composite skin. Marked locations on the skin surface can be abraded and cleaned prior to bonding the stringer standoff mounts onto the inner surface of the skin.

(59) Alternatively, the composite skin can be made from “dry” fabric/fiber and using a resin infusion process or it can be made from “wet” prepreg fabric/fibers. The composite skin can be cured using any of, but not limited to, the following techniques and/or methods or any derivatives thereof: room temperature cure, oven temperature cure, autoclave cure and electron beam cure.

(60) The composite skins can be manufactured using any of, but not limited to, the following tooling methods for composite fabrication or any derivatives thereof: bagging (vacuum or autoclave) on male or female tools, and clam-shell molds.

(61) The composite skin can be manufactured from any combination of, but not limited to, the following materials or any derivatives thereof: carbon fiber, fiberglass, Kevlar, carbon fiber nanotubes, other composite fabric/fiber materials (synthetic and natural), polymers, ceramics, metallic foils, and thin metallic sheet.

(62) The stringer standoff mounts can be manufactured in any of a variety of ways. One exemplary embodiment includes stereolithography 3D printing. Rigid UV-cured plastic standoff mounts can be reinforced by bonding a metallic U-shaped cap onto the outside surface of each stringer standoff mount, which helps to provide an optimal strength to weight ratio. Standoff mounts can also include a knurled or cross-hatched pattern embossed into the underside of the base for increasing surface area for bonding and ensuring a consistent bond line thickness.

(63) The stringer standoff mounts can be abraded, cleaned and then bonded into place on the inner skin surface of the flexible composite with a high-strength two-part epoxy designed specifically for bonding carbon fiber composites. The stringer standoff mounts can be made from any combination of, but not limited to, the following materials or any derivatives thereof: metals, polymers, ceramics, plastics, composite fiber/fabric, and chopped composite fiber.

(64) The stringer standoff mounts can be manufactured using any combination of the following techniques or any derivatives thereof: 3D printing, machining, casting, forging, stamping, injection molding, composite fabrication, and so forth.

(65) The overall design is configured so that the flexible stringers are inserted and threaded through the openings of the corresponding stringer standoff mounts, as well as, attaching to (and/or gliding on, and/or being inserted through) any additional active and passive structural components. The stringers may be formed into a variety of cross-sectional shapes, including but not limited to circular, square, rectangular, polygonal and oval, I-beam, C-channel and T-beam. Further, the stringers may be solid or hollow depending on the application. Stringers be manufactured using any combination of fiber/fabric directions from any composite material and/or metallic, ceramic, polymer material in any form including but not limited to sheet/film. The skin configurations can be tailored by adjusting the quantity, spacing, size, locations and material properties of the flexible stringers and the stringer standoff mounts.

(66) The actuator ribs and compliant support ribs can be manufactured using any combination of the following techniques, or any derivatives thereof: 3D printing, machining, casting, forging, stamping, injection molding, and composite fabrication. Material selections for the compliant

support ribs can be made from any combination of metals, plastics, ceramics, polymers, composite fiber/fabric, and chopped composite fiber.

(67) In one aspect the present disclosure includes a morphing structure comprising: an underlying structure having actively controlled elements and passive elements; a non-stretchable flexible skin having an inner surface and an outer surface connected to the underlying structure to form an airfoil with a pressure side surface and a suction side surface; and a control system operable to change the camber of the airfoil.

(68) In refining aspects, the flexible skin is substantially continuous from a leading edge to a trailing edge of the airfoil on the suction side surface; the control system is operable to twist or bend the skin without changing a length or width of the skin; further comprising an actuator to move the actively controlled structure and morph the skin to change the shape of the airfoil; wherein morphing the skin causes a lift distribution of the airfoil to change without separate movable control surfaces; and wherein the skin is formed from a composite material having anisotropic properties made from a plurality of unidirectional fabrics.

(69) In another aspect, the present disclosure includes a vehicle with a morphing airfoil comprising: a plurality of actuator ribs configured to move under an actuation force; non-actuated support structure positioned between the actuator ribs; and a non-stretchable flexible skin covering the actuator ribs and the non-actuated support structure formed in a shape of an airfoil.

(70) In refining aspects, the non-stretchable flexible skin forms a substantially continuous surface across a portion of the airfoil; wherein the vehicle is an aircraft; wherein the vehicle is an automobile; further comprising an actuator connected to the actuator ribs operable for moving the trailing edge thereof; wherein the flexible skin is bendable in all directions; wherein the flexible skin does not expand or contract along a length or width thereof when morphed into a different shape; wherein the flexible skin is formed from a plurality of layers of composite material; wherein a bottom portion of the actuator rib is movable relative to a top portion of the actuator rib to cause the trailing edge of the airfoil to deflect; and wherein the skin is slidably connected to the non-actuated support structure; wherein the skin is slidably connected to the actuator ribs.

(71) In another aspect, the present disclosure includes a method of morphing an airfoil comprising: deflecting a trailing edge of an actuator rib; morphing a non-stretchable flexible skin in response to the deflecting of the actuator rib; and moving a trailing edge of a non-actuated support structure in response to morphing the skin.

(72) In refining aspects, the method includes changing a lift distribution on the airfoil in response to the morphing of the skin; wherein the skin is substantially continuous along a suction side of the airfoil; wherein the morphing includes bending and torsional twisting of the skin along a length thereof; wherein the moving includes nonlinear deflection of the trailing edge; and wherein the skin is constrained by stringers slidably coupled to slots formed with actuator ribs and the non-actuated support structure.

(73) While the invention has been illustrated and described in detail in the drawings and foregoing description, the same is to be considered as illustrative and not restrictive in character, it being understood that only the preferred embodiments have been shown and described and that all changes and modifications that come within the spirit of the inventions are desired to be protected. It should be understood that while the use of words such as preferable, preferably, preferred or more preferred utilized in the description above indicate that the feature so described may be more desirable, it nonetheless may not be necessary and embodiments lacking the same may be contemplated as within the scope of the invention, the scope being defined by the claims that follow. In reading the claims, it is intended that when words such as “a,” “an,” “at least one,” or “at least one portion” are used there is no intention to limit the claim to only one item unless specifically stated to the contrary in the claim. When the language “at least a portion” and/or “a portion” is used the item can include a portion and/or the entire item unless specifically stated to the contrary.

(74) Unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings. Further, “connected” and “coupled” are not restricted to physical or mechanical connections or couplings.

Claims

1. A morphing structure comprising: an underlying structure having actively controlled elements and passive elements; a non-stretchable flexible skin having an inner surface and an outer surface connected to the underlying structure to form an airfoil with a pressure side surface and a suction side surface; a control system operable to change the camber of the airfoil; wherein the actively controlled elements extend forward of a spar toward a leading edge and back toward the trailing edge of the airfoil to control the camber of the flexible skin; wherein the actively controlled elements include a compliant mechanism having a plurality of vertical compliant flexible legs spaced apart from one another and configured to permit a lower surface of an actuator rib to move relative to an upper surface of the actuator rib; wherein the passively controlled elements extend from the spar back to the trailing edge of the airfoil and are positioned between the actively controlled elements; and wherein the flexible skin controls the camber shape of the passive elements.
 2. The morphing structure of claim 1, wherein the flexible skin is substantially continuous from a leading edge to a trailing edge of the airfoil on the suction side surface.
 3. The morphing structure of claim 1, wherein the control system is operable to twist or bend the skin without changing a length or width of the skin.
 4. The morphing structure of claim 1, further comprising an actuator to move the actively controlled structure and morph the skin to change the shape of the airfoil.
 5. The morphing structure of claim 4, wherein morphing the skin causes a lift distribution of the airfoil to change without separate movable control surfaces.
 6. The morphing structure of claim 1, wherein the skin is formed from a composite material having anisotropic properties made from a plurality of unidirectional fabrics.
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